BitFlip

CloudSherpa

Software Requirements Specification

Contents

1 | Introduction

CloudSherpa is a multi-cloud cost management and optimization platform designed to address the growing complexity of managing cloud infrastructure across AWS, Azure, and Google Cloud Platform.

As organizations increasingly adopt multi-cloud strategies, they face challenges in monitoring spending, identifying cost anomalies, and optimizing resource utilization across fragmented platforms. CloudSherpa consolidates billing and usage data from all three major cloud providers into a unified dashboard, providing real-time cost visibility, predictive forecasting, and actionable optimization recommendations.

The platform enables teams to make data-driven decisions, reduce cloud expenditure, and improve resource efficiency through comprehensive analytics, automated alerts, and scenario planning capabilities.

2 | User Stories / User Characteristics

2.1 User Characteristics

To adequately support CloudSherpa's multi-cloud optimization platform, the system targets three primary user groups.

- **Platform Administrator / IT User:**
  - *Characteristics:* High technical expertise, familiar with cloud infrastructure (AWS, Azure, GCP), IAM roles, and system administration.
  - *System Usage:* Responsible for configuring the platform, connecting/revoking cloud provider credentials, managing role-based access control, and ensuring data ingestion pipelines are functioning.
- **Finance-Focused User / Management:**
  - *Characteristics:* Low to medium technical expertise; high financial and business acumen. Prioritizes plain-language summaries over technical jargon.

  - *System Usage:* Utilizes the platform to view forecasted cloud costs, set budget threshold alerts, generate downloadable reports, and track estimated savings from optimization recommendations.

- **Technical Lead / Operations Manager:**
  - *Characteristics:* High technical expertise; responsible for architecture, deployments, and infrastructure efficiency.
  - *System Usage:* Monitors cross-provider resource usage, surfaces unusual trend shifts, identifies idle resources, and applies right-sizing recommendations.

2.2 User Stories

2.2.1 Platform Administrator / IT User Stories

- **US1:** As a platform user, I want to connect my AWS, Azure, and GCP accounts so that CloudSherpa can collect usage, billing, and performance data.
- **US2:** As a platform administrator, I want provider-specific ingestion jobs to run automatically so that data remains up to date without manual effort.
- **US3:** As a platform user, I want failed cloud integrations to show clear error feedback so that I can fix configuration issues quickly.
- **US4:** As a data processing service, I want incoming provider data to be normalized so that analytics can be performed consistently across clouds.
- **US5:** As a system administrator, I want processed data pipelines to be reliable so that dashboards and services reflect trusted information.
- **US6:** As a new user, I want to create an account and log in securely so that I can access CloudSherpa safely.
- **US7:** As a platform user, I want to link multiple cloud environments to my profile so that I can manage them from one place.
- **US8:** As an administrator, I want role-based access control so that users only see what they are authorized to access.
- **US9:** As a platform user, I want to manage and revoke connected cloud accounts so that I stay in control of my integrations.

2.2.2 Finance-Focused User / Management Stories

- **US10:** As a finance-focused user, I want forecasted cloud costs for upcoming periods so that I can plan budgets proactively.
- **US11:** As a platform user, I want to see spending trends over time so that I can identify whether my costs are rising or falling.
- **US12:** As a platform user, I want to define budget thresholds so that I am notified before overspending becomes severe.
- **US13:** As a manager, I want downloadable reports so that I can share cloud-cost insights with stakeholders.
- **US14:** As a non-technical user, I want the dashboard to present information clearly so that I can still make informed decisions.
- **US15:** As a platform user, I want trend analysis summaries so that I can understand long-term cost behavior.
- **US16:** As an auditor or reviewer, I want visual summaries of key metrics so that I can assess cloud usage efficiently.
- **US17:** As a platform user, I want reports filtered by provider, time period, or environment so that they are relevant to my use case.

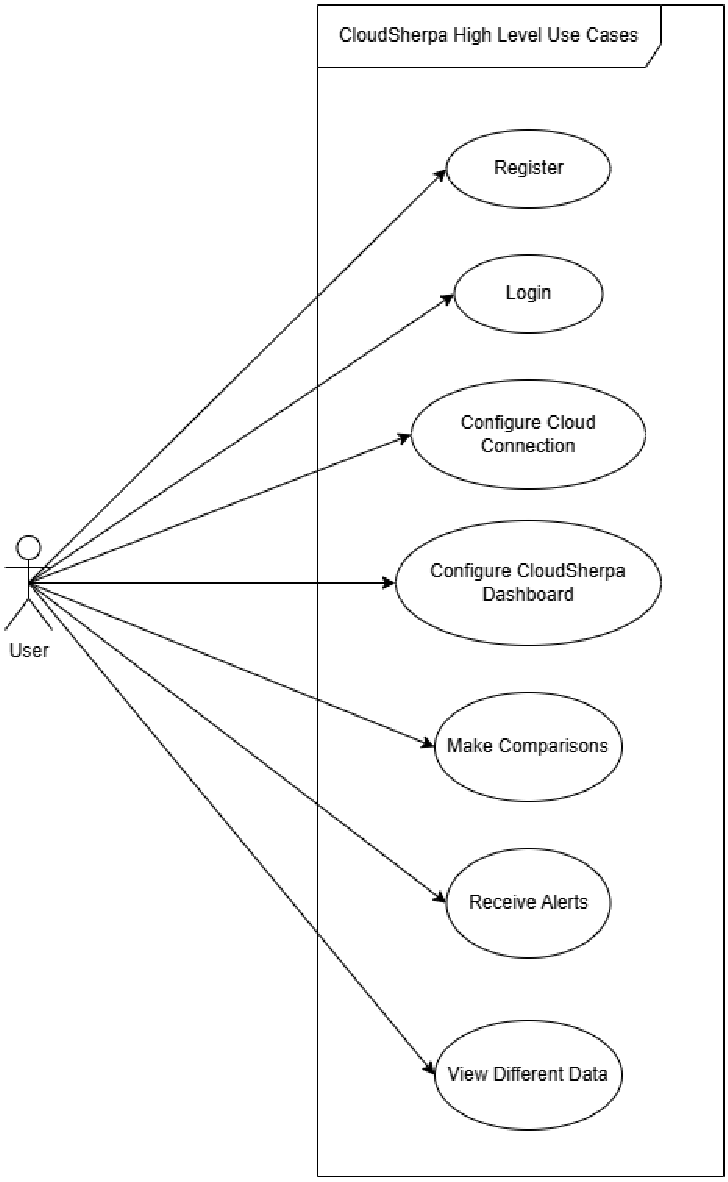
2.2.3 Technical Lead / Operations Manager Stories

- **US18:** As an operations manager, I want forecasts per provider so that I can understand where future spend will be concentrated.
- **US19:** As a platform user, I want the system to surface unusual trend shifts so that I can investigate them early.
- **US20:** As a platform user, I want the system to identify idle resources so that I can reduce unnecessary cloud spend.
- **US21:** As a platform user, I want recommendations for overprovisioned resources so that I can right-size infrastructure.
- **US22:** As a technical decision-maker, I want each recommendation to include estimated savings so that I can prioritize actions.
- **US23:** As a platform user, I want recommendations to be grouped by severity so that I can focus on the highest-value optimizations first.
- **US24:** As a platform user, I want a dashboard overview of cloud usage and spend so that I can understand my environment at a glance.

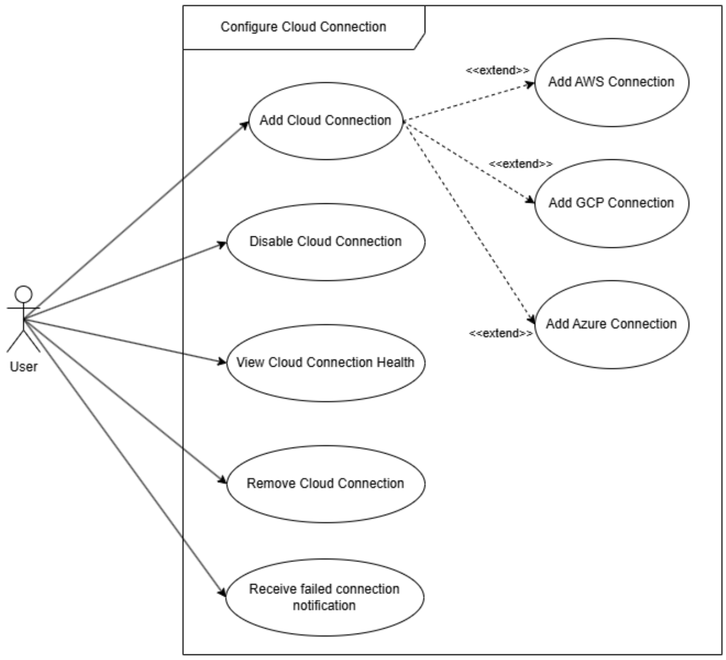
- **US25:** As a platform user, I want visual charts for forecasts and trends so that I can interpret data quickly.
- **US26:** As a platform user, I want recommendations and anomalies displayed in the dashboard so that I can take action from one place.
- **US27:** As a platform user, I want to receive alerts when anomalous spending is detected so that I can respond quickly.
- **US28:** As a platform user, I want optimization opportunities to trigger notifications so that I do not miss savings.
- **US29:** As a platform user, I want alerts delivered through channels like email or SMS so that I can receive them where convenient.
- **US30:** As a platform user, I want to compare costs across AWS, Azure, and GCP so that I can identify the most cost-effective provider.
- **US31:** As a platform user, I want to compare resource usage across providers so that I can understand efficiency differences.
- **US32:** As a technical lead, I want side-by-side provider views so that I can support architecture and deployment decisions.
- **US33:** As a manager, I want comparison summaries over time so that I can track which providers are becoming more or less economical.

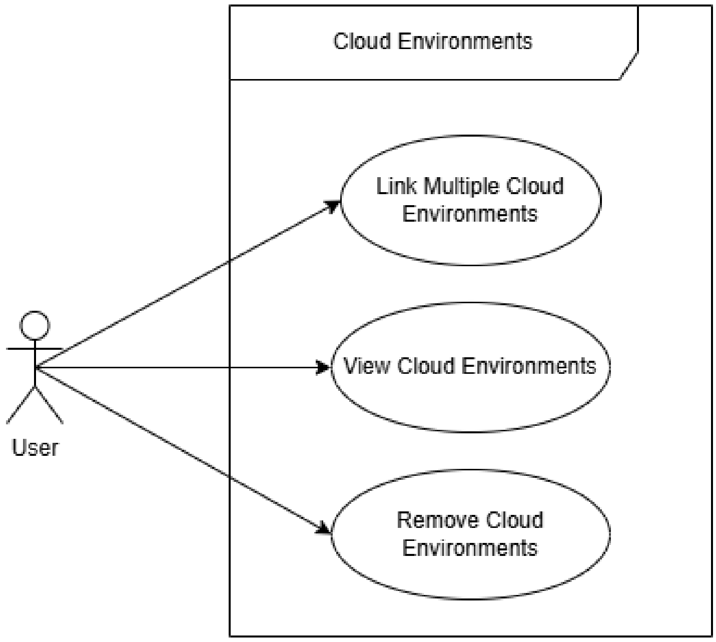
3 | Use Case Diagrams

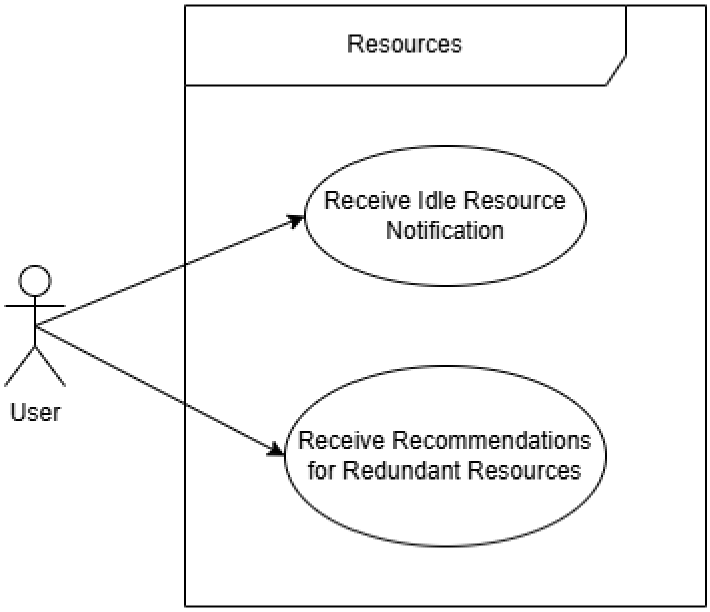
3.1 High-Level Use Case Diagram

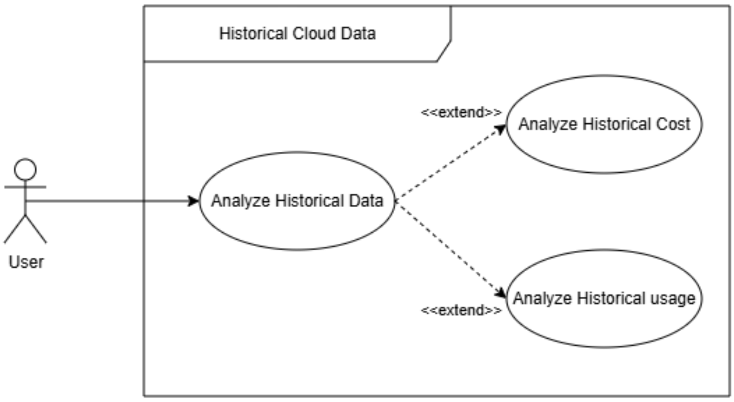


3.2 Individual Use Case Diagrams






Resources
Receive Idle Resource Notification
Receive Recommendations for Redundant Resources
User


Historical Cloud Data
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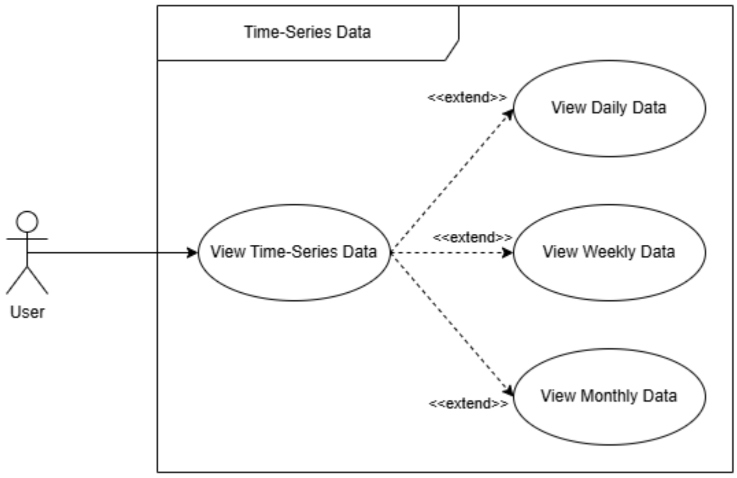
Analyze Historical Cost
Analyze Historical Data
User
Analyze Historical usage
<<extend>>


Time-Series Data
User
View Time-Series Data
<<extend>>
View Daily Data
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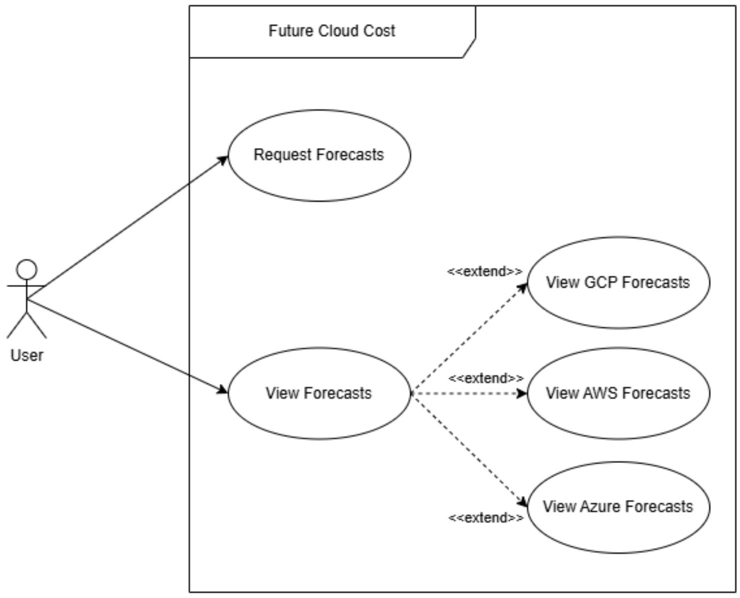
View Weekly Data
<<extend>>
View Monthly Data


Future Cloud Cost
User
Request Forecasts
View Forecasts
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View GCP Forecasts
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View AWS Forecasts
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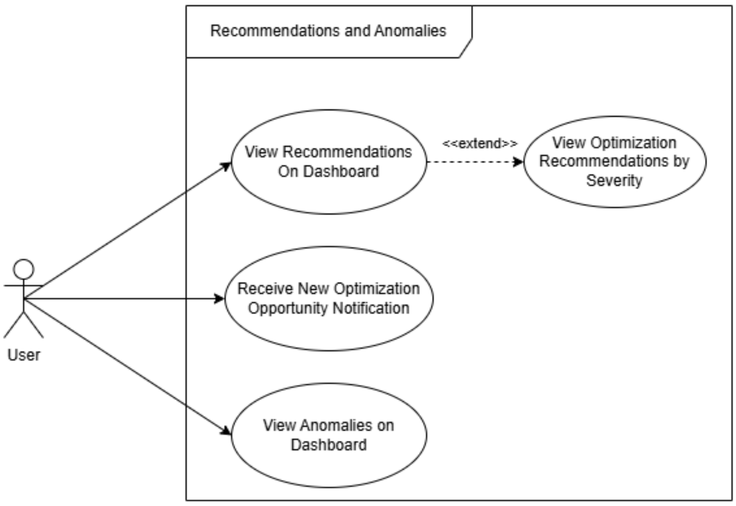
View Azure Forecasts


Recommendations and Anomalies
View Recommendations On Dashboard
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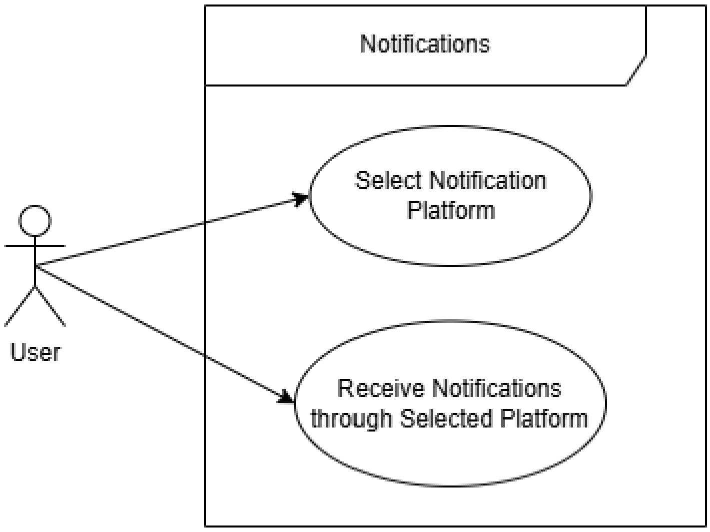
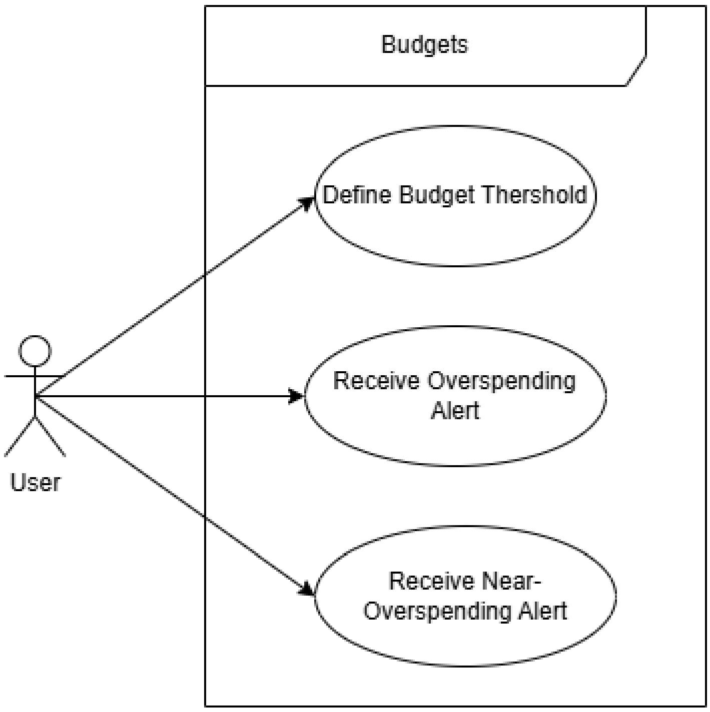
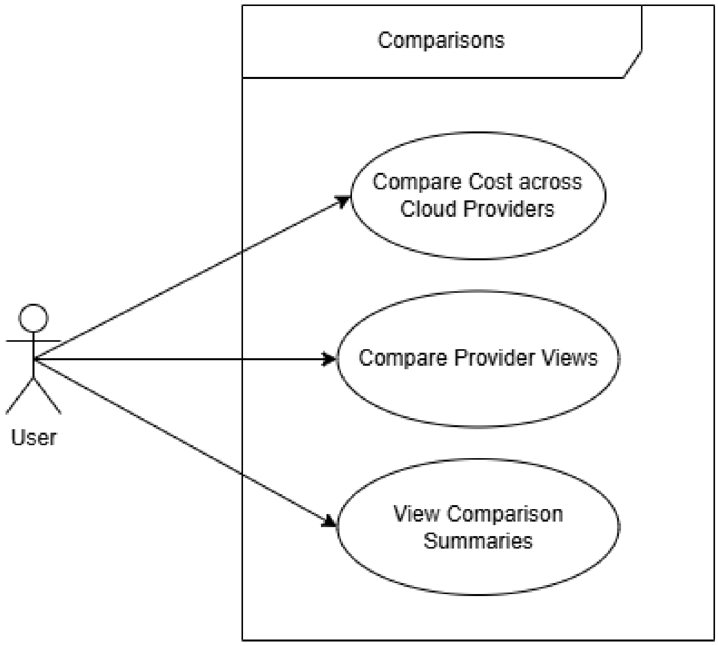
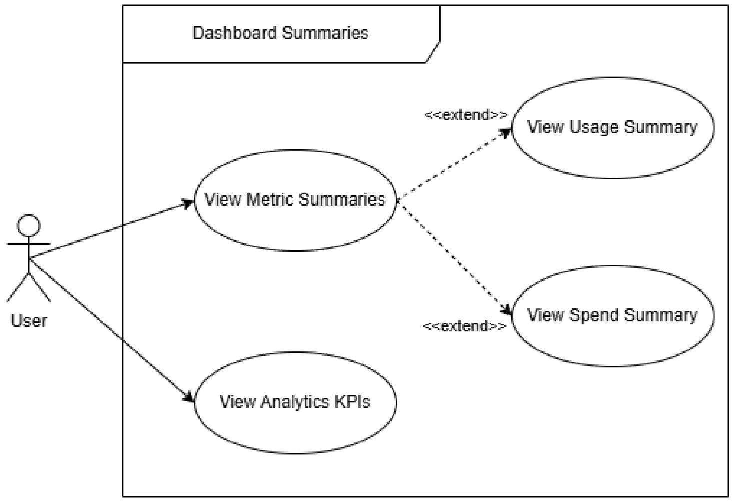
View Optimization Recommendations by Severity
Receive New Optimization Opportunity Notification
View Anomalies on Dashboard
User


Notifications
Select Notification Platform
Receive Notifications through Selected Platform
User


Budgets
Define Budget Thershold
Receive Overspending Alert
Receive Near-Overspending Alert
User


Comparisons
Compare Cost across Cloud Providers
Compare Provider Views
View Comparison Summaries
User


Dashboard Summaries
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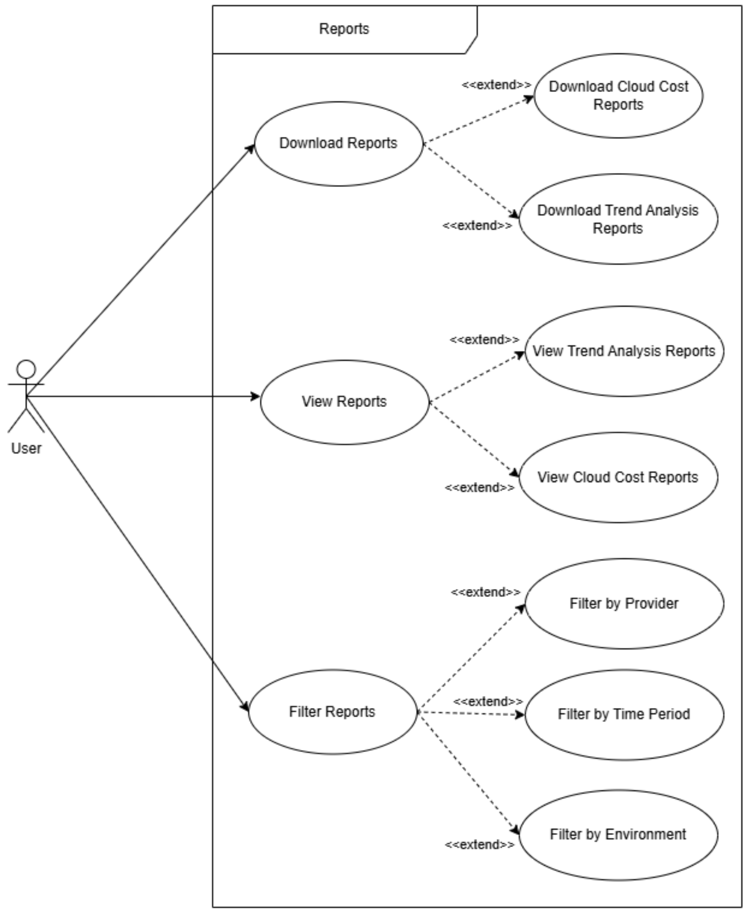
View Usage Summary
View Metric Summaries
User
View Spend Summary
<<extend>>
View Analytics KPIs


Reports
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Download Cloud Cost Reports
Download Reports
Download Trend Analysis Reports
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View Trend Analysis Reports
User
View Reports
View Cloud Cost Reports
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Filter by Provider
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Filter Reports
Filter by Time Period
Filter by Environment
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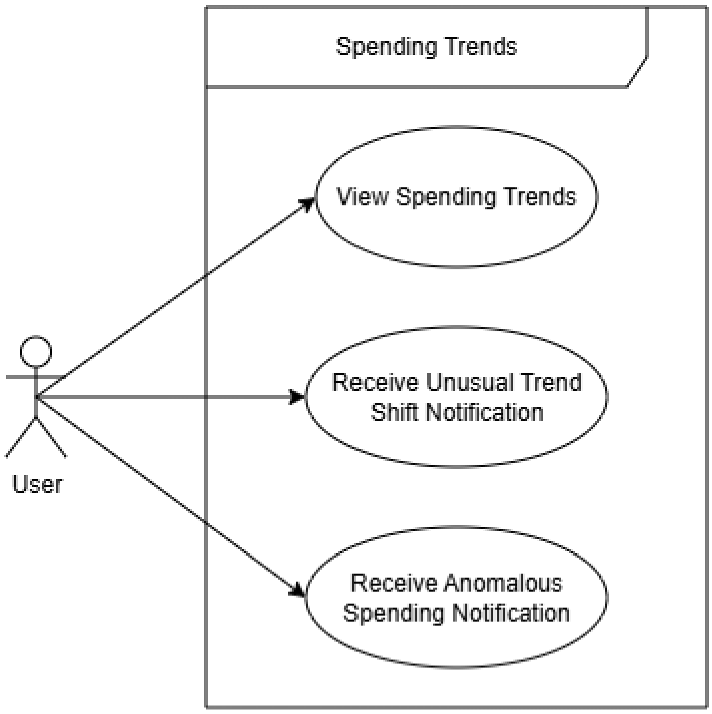

4 Functional Requirements

4.1 Cloud Data Integration

FR1: Cloud Data Integration *(Subsystem: Ingestion Service - Connection Manager and Cloud Connector Modules)*

- **FR1.1:** The system shall integrate with cloud providers (AWS, Azure, and GCP).
  - **FR1.1.1** The system shall integrate with the AWS Cloud Provider
    - **FR 1.1.1.1** The system shall ingest metrics from AWS
    - **FR 1.1.1.2** The system shall ingest billing data from AWS
- **FR1.2:** The system shall collect usage metrics, billing data, and performance data needed for analysis.

4.2 Data Storage and Processing

FR2: Data Storage and Processing *(Subsystem: Ingestion Service - Normalization module and Persistence Service)*

- **FR2.1:** The system shall store historical cloud data.
- **FR2.2:** The system shall process and aggregate time-series data to prepare it for analytical workloads.

4.3 Cost Prediction

FR3: Cost Prediction *(Subsystem: Application Service - Forecast module)*

- **FR3.1:** The system shall apply machine learning models to forecast future cloud costs.
- **FR3.2:** The system shall identify and provide forward-looking insights into spending trends.

4.4 Optimization Recommendations

FR4: Optimization Recommendations *(Subsystem: Application Service - Optimization module)*

- **FR4.1:** The system shall detect inefficiencies in infrastructure utilization, such as idle and overprovisioned resources.
- **FR4.2:** The system shall provide actionable cost-saving and corrective recommendations based on detected inefficiencies.

4.5 Web-Based Dashboard

FR5: Web-Based Dashboard *(Subsystem: Presentation Service - Dashboard module)*

- **FR5.1:** The system shall provide a unified web interface to visualize usage metrics.
- **FR5.2:** The system shall display cost forecasts and present optimization recommendations.

4.6 User and Account Management

FR6: User and Account Management *(Subsystem: Application Service - Authentication module)*

- **FR6.1:** The system shall allow users to connect multiple cloud accounts and manage environments.
- **FR6.2:** The system shall control access permissions and manage user identities securely.

4.7 Alerts and Notifications

FR7: Alerts and Notifications *(Subsystem: Application Service - Anomaly and Alert modules)*

- **FR7.1:** The system shall proactively notify users of detected cloud-cost anomalies.
- **FR7.2:** The system shall alert users when configured budget thresholds are reached.
- **FR7.3:** The system shall send notifications via user-selected channels (e.g., email or SMS).

4.8 Multi-Cloud Cost Comparison

FR8: Multi-Cloud Cost Comparison *(Subsystem: Application Service - Analytics and optimization modules)*

- **FR8.1:** The system shall support direct comparison of costs across supported cloud providers.
- **FR8.2:** The system shall compare resource usage metrics across providers to help identify cost-effective environments.

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5 | Non-Functional Requirements

5.1 Performance

NFR1.1: The system should load the main dashboard within 2 seconds and update visualization widgets within 1 second of a user modifying a time window, while seamlessly handling the throughput required to ingest metrics from 1,000 live deployments.

NFR1.2: The ingestion pipeline shall process, normalize, and aggregate 100,000 raw billing and usage records from AWS, Azure, and GCP APIs into the unified internal schema within 60 seconds of data retrieval.

5.2 Scalability

NFR2.1: The system should successfully support 500 concurrent dashboard users and manage 1,000 active cloud connections without experiencing more than a 10% degradation in dashboard response times.

NFR2.2: The time-series database architecture shall scale dynamically to accommodate a 200% year-over-year growth in stored historical cloud data without requiring manual partitioning or major architectural changes.

5.3 Security

NFR3.1: All sensitive user data and stored cloud provider credentials must be encrypted using TLS 1.3 in transit, and the system must enforce Multi-Factor Authentication (MFA) on 100% of accounts.

NFR3.2: The system shall utilize secure, least-privilege identity access management (IAM) strategies, ensuring that CloudSherpa holds read-only access to client AWS, Azure, and GCP billing APIs and ensure that none of CloudSherpa's requests require write permissions.

5.4 Reliability

NFR4.1: The system should achieve 99.9% uptime for both the dashboard and the metrics ingestion pipeline.

5.5 Maintainability

NFR5.1: The codebase must maintain a minimum of 80% automated test coverage across all core components, and the continuous integration pipeline should enable new feature deployments or bug fixes to reach production within 2 hours.

NFR5.2: The data normalization module must be sufficiently decoupled so that integrating a new cloud provider API, or updating an existing provider's changed billing schema, requires no more than 40 hours of developer effort.

5.6 Usability

NFR6.1: A new non-technical user should be able to complete core tasks (e.g., viewing forecasts or checking anomaly alerts) within 5 minutes of first using the system, achieving at least 85% user satisfaction during usability testing.

NFR6.2: The CloudSherpa dashboard shall render correctly on modern web browsers to ensure plain-language summaries remain highly accessible.

5.7 Availability

NFR7.1: The system should be available 24/7, excluding scheduled maintenance periods not exceeding 2 hours per month.

NFR7.2: The web dashboard and historical cost databases shall remain 100% available and accessible to users even if one or more underlying cloud provider APIs (AWS, Azure, GCP) experience a temporary outage.

6 | Domain Model

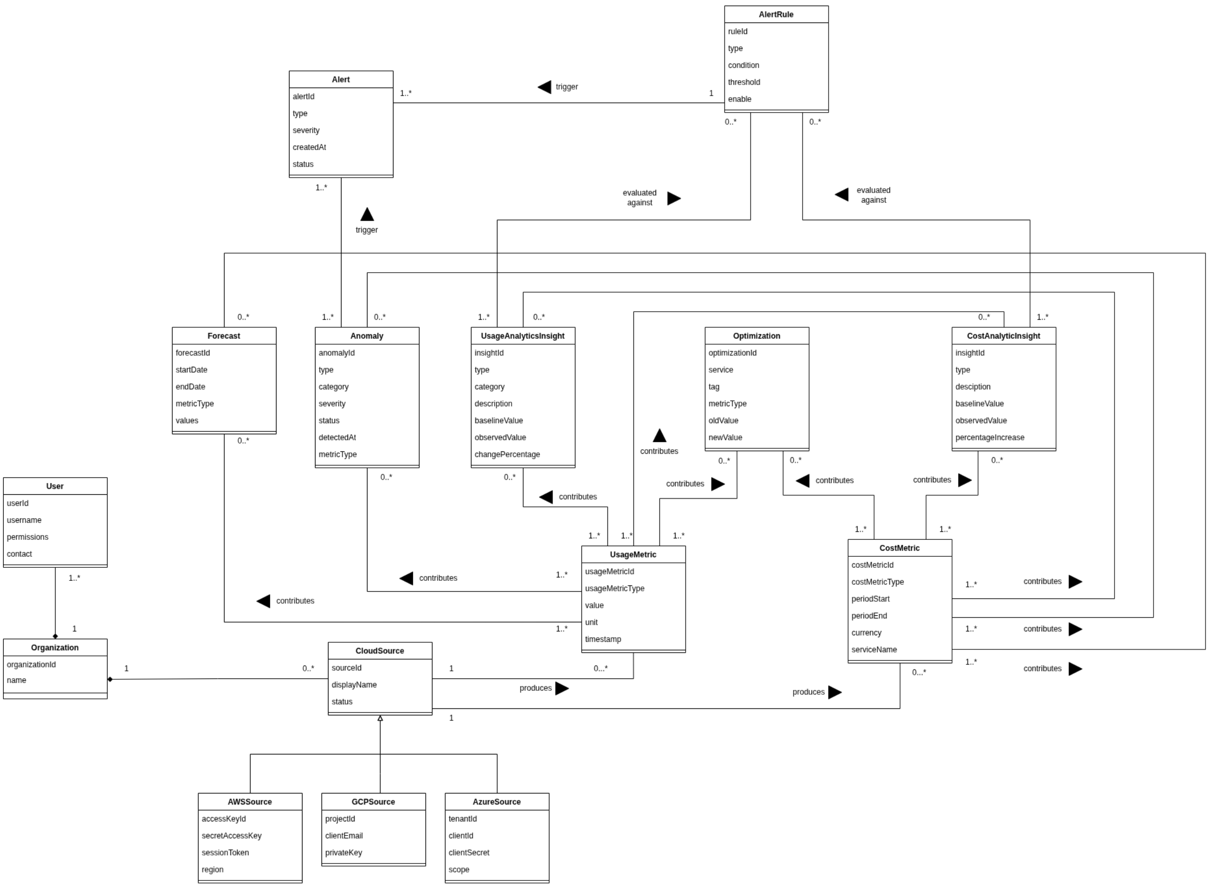

Figure 1: CloudSherpa Domain Model